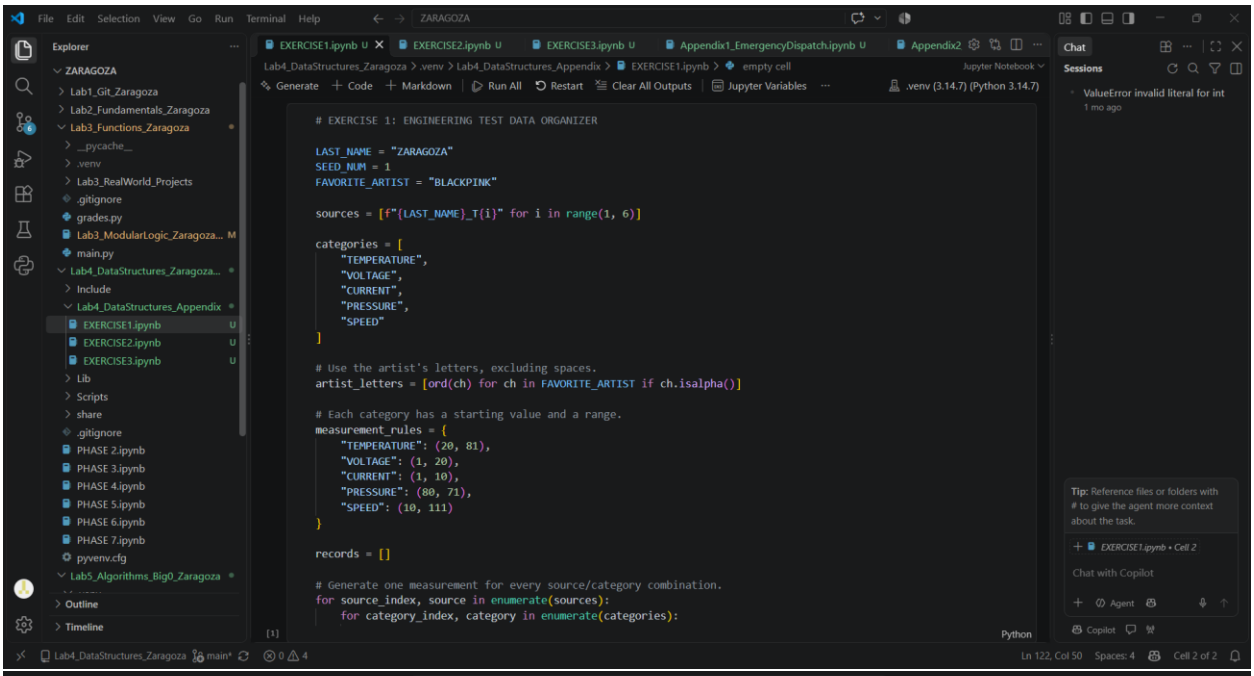
**Zaragoza, Samson Jr. C.** **Computer Programming**

**BSECE – 1C** **Activity 4: Appendix**

## # EXERCISE 1: Engineering Test Data Organizer



```
# EXERCISE 1: ENGINEERING TEST DATA ORGANIZER

LAST_NAME = "ZARAGOZA"
SEED_NUM = 1
FAVORITE_ARTIST = "BLACKPINK"

sources = [f"{LAST_NAME}_T{i}" for i in range(1, 6)]

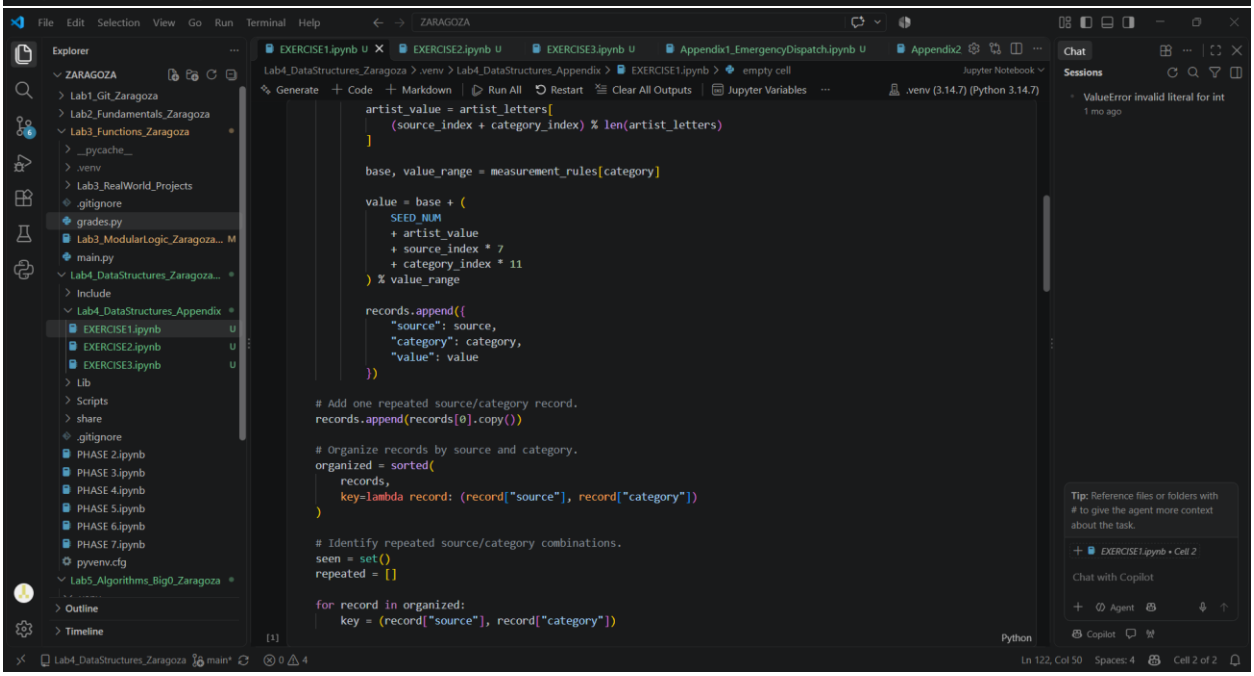
categories = [
    "TEMPERATURE",
    "VOLTAGE",
    "CURRENT",
    "PRESSURE",
    "SPEED"
]

# Use the artist's letters, excluding spaces.
artist_letters = [ord(ch) for ch in FAVORITE_ARTIST if ch.isalpha()]

# Each category has a starting value and a range.
measurement_rules = {
    "TEMPERATURE": (20, 81),
    "VOLTAGE": (1, 20),
    "CURRENT": (1, 10),
    "PRESSURE": (80, 71),
    "SPEED": (10, 111)
}

records = []

# Generate one measurement for every source/category combination.
for source_index, source in enumerate(sources):
    for category_index, category in enumerate(categories):
```



```
            artist_value = artist_letters[
                (source_index + category_index) % len(artist_letters)
            ]

            base, value_range = measurement_rules[category]

            value = base + (
                SEED_NUM
                + artist_value
                + source_index * 7
                + category_index * 11
            ) % value_range

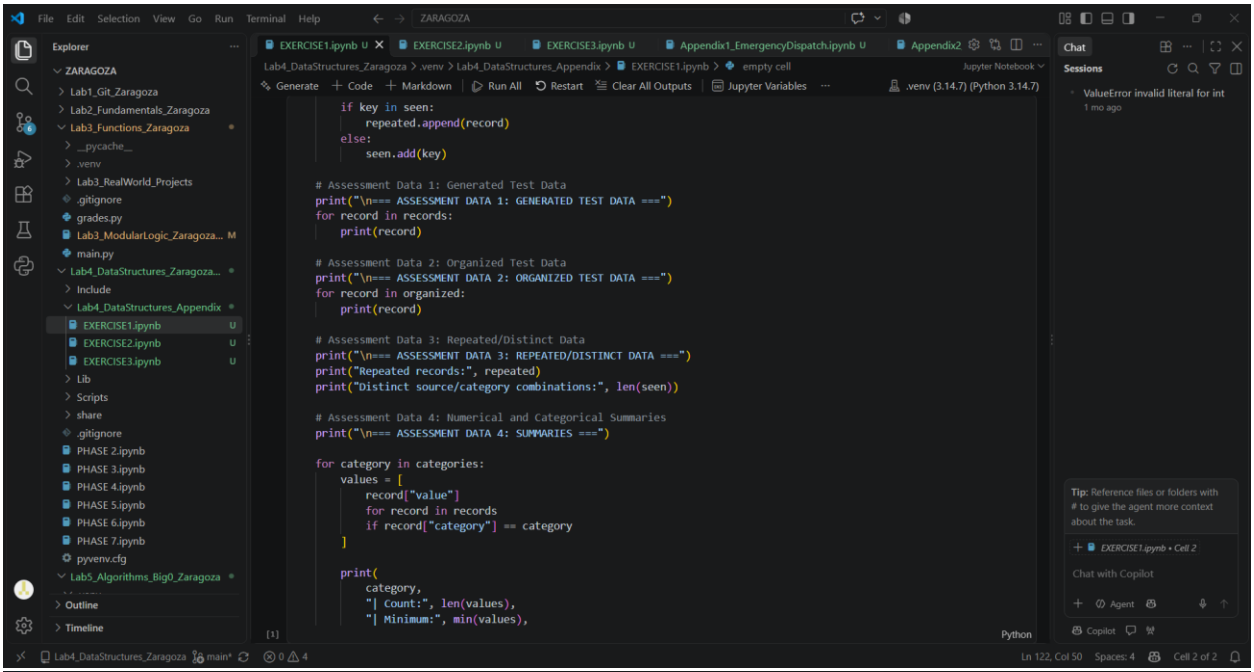
            records.append({
                "source": source,
                "category": category,
                "value": value
            })

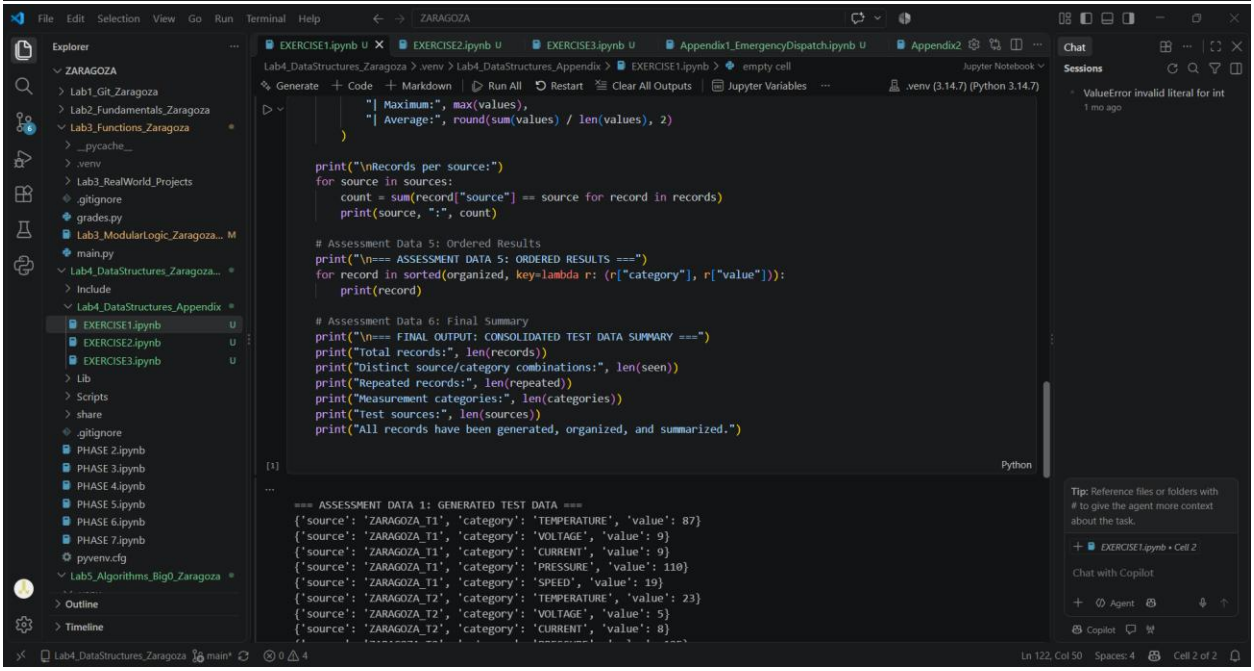
# Add one repeated source/category record.
records.append(records[0].copy())

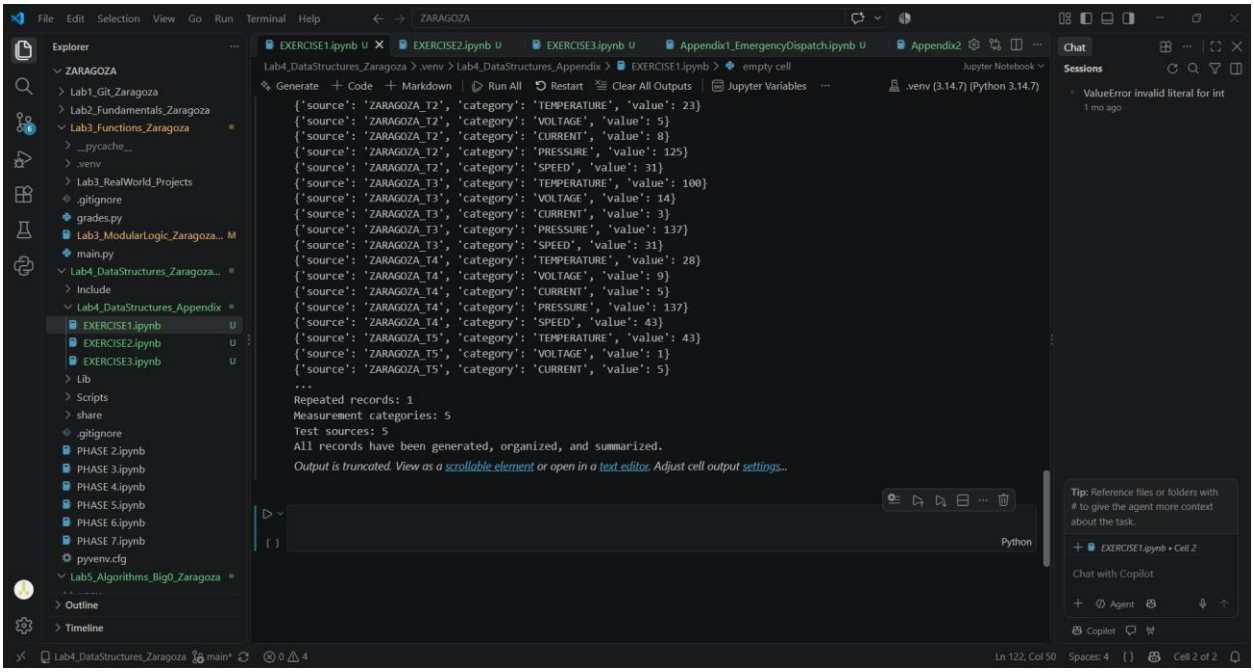
# Organize records by source and category.
organized = sorted(
    records,
    key=lambda record: (record["source"], record["category"])
)

# Identify repeated source/category combinations.
seen = set()
repeated = []

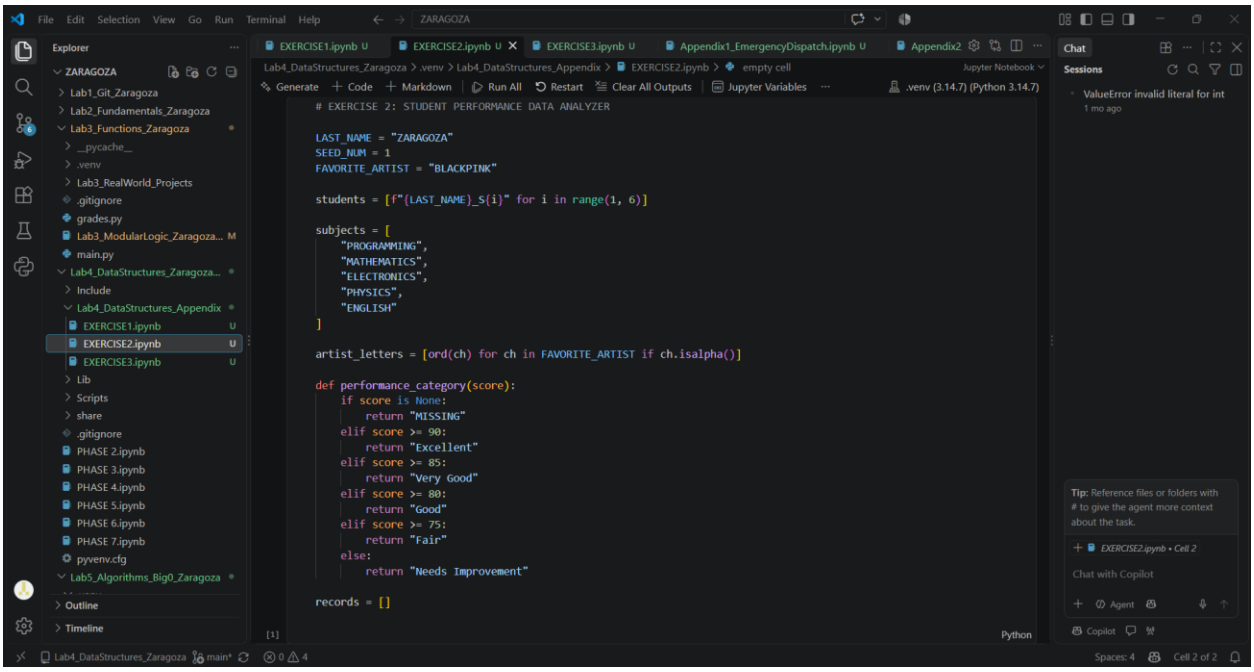
for record in organized:
    key = (record["source"], record["category"])
```

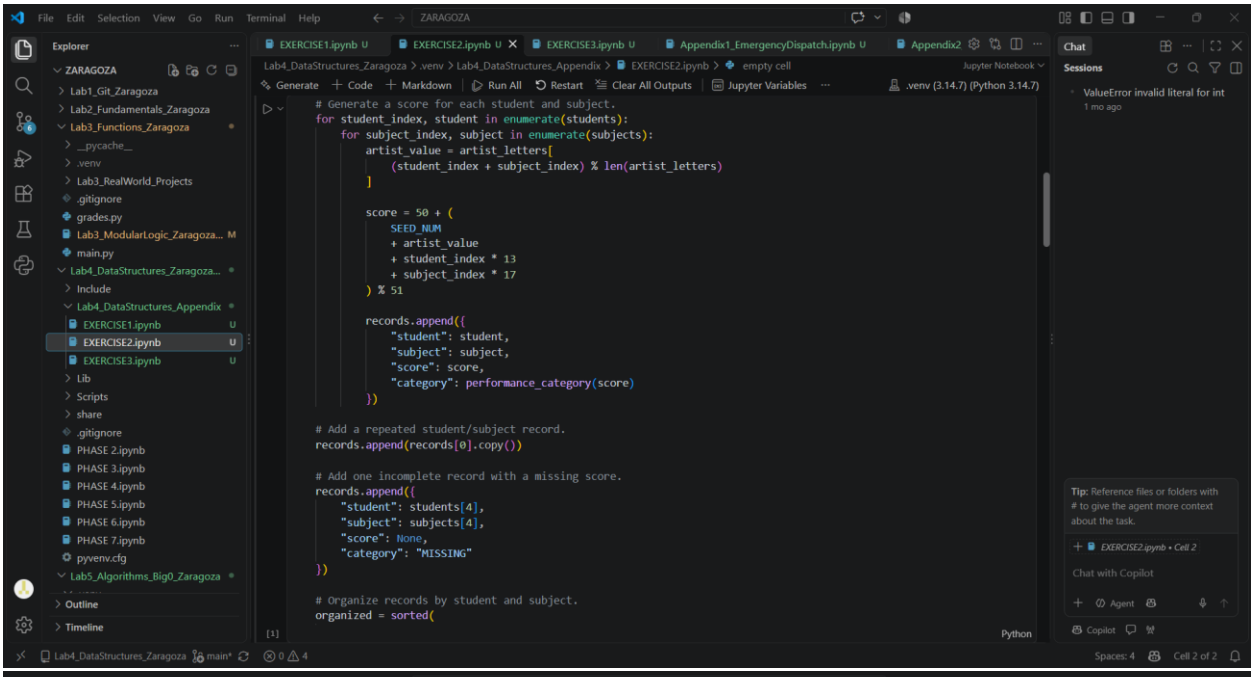

File Edit Selection View Go Run Terminal Help
ZARAGOZA
Explorer
ZARAGOZA
Lab1_Git_Zaragoza
Lab2_Fundamentals_Zaragoza
Lab3_Functions_Zaragoza
__pycache__
.venv
Lab3_RealWorld_Projects
.gitignore
grades.py
Lab3_ModularLogic_Zaragoza... M
main.py
Lab4_DataStructures_Zaragoza...
Include
Lab4_DataStructures_Appendix
EXERCISE1.ipynb U
EXERCISE2.ipynb U
EXERCISE3.ipynb U
Lib
Scripts
share
.gitignore
PHASE 2.ipynb
PHASE 3.ipynb
PHASE 4.ipynb
PHASE 5.ipynb
PHASE 6.ipynb
PHASE 7.ipynb
pyvenv.cfg
Lab5_Algorithms_Big0_Zaragoza
Outline
Timeline
EXERCISE1.ipynb U
EXERCISE2.ipynb U
EXERCISE3.ipynb U
Appendix1_EmergencyDispatch.ipynb U
Appendix2
Lab4_DataStructures_Zaragoza > .venv > Lab4_DataStructures_Appendix > EXERCISE1.ipynb > empty cell
Jupyter Notebook
Generate
Code
Markdown
Run All
Restart
Clear All Outputs
Jupyter Variables
.venv (3.14.7) (Python 3.14.7)
if key in seen:
repeated.append(record)
else:
seen.add(key)
# Assessment Data 1: Generated Test Data
print("\n=== ASSESSMENT DATA 1: GENERATED TEST DATA ===")
for record in records:
print(record)
# Assessment Data 2: Organized Test Data
print("\n=== ASSESSMENT DATA 2: ORGANIZED TEST DATA ===")
for record in organized:
print(record)
# Assessment Data 3: Repeated/Distinct Data
print("\n=== ASSESSMENT DATA 3: REPEATED/DISTINCT DATA ===")
print("Repeated records:", repeated)
print("Distinct source/category combinations:", len(seen))
# Assessment Data 4: Numerical and Categorical Summaries
print("\n=== ASSESSMENT DATA 4: SUMMARIES ===")
for category in categories:
values = [
record["value"]
for record in records
if record["category"] == category
]
print(
category,
"| Count:", len(values),
"| Minimum:", min(values),
[1]
Python
Chat
Sessions
ValueError invalid literal for int
1 mo ago
Tip: Reference files or folders with # to give the agent more context about the task.
EXERCISE1.ipynb • Cell 2
Chat with Copilot
Agent
Copilot
Lab4_DataStructures_Zaragoza main*
0 4
Ln 122, Col 50
Spaces: 4
Cell 2 of 2

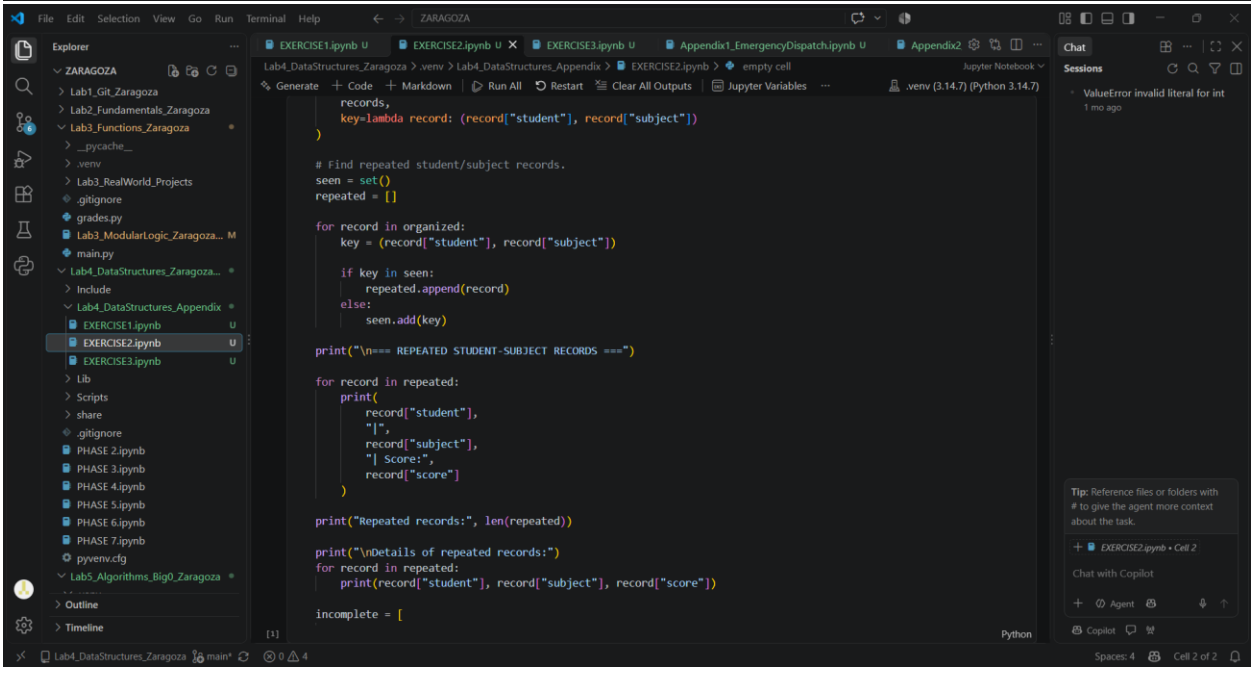

File Edit Selection View Go Run Terminal Help
ZARAGOZA
Explorer
ZARAGOZA
Lab1_Git_Zaragoza
Lab2_Fundamentals_Zaragoza
Lab3_Functions_Zaragoza
__pycache__
.venv
Lab3_RealWorld_Projects
.gitignore
grades.py
Lab3_ModularLogic_Zaragoza... M
main.py
Lab4_DataStructures_Zaragoza...
Include
Lab4_DataStructures_Appendix
EXERCISE1.ipynb U
EXERCISE2.ipynb U
EXERCISE3.ipynb U
Lib
Scripts
share
.gitignore
PHASE 2.ipynb
PHASE 3.ipynb
PHASE 4.ipynb
PHASE 5.ipynb
PHASE 6.ipynb
PHASE 7.ipynb
pyvenv.cfg
Lab5_Algorithms_Big0_Zaragoza
Outline
Timeline
EXERCISE1.ipynb U
EXERCISE2.ipynb U
EXERCISE3.ipynb U
Appendix1_EmergencyDispatch.ipynb U
Appendix2
Lab4_DataStructures_Zaragoza > .venv > Lab4_DataStructures_Appendix > EXERCISE1.ipynb > empty cell
Jupyter Notebook
Generate
Code
Markdown
Run All
Restart
Clear All Outputs
Jupyter Variables
.venv (3.14.7) (Python 3.14.7)
"| Maximum:", max(values),
"| Average:", round(sum(values) / len(values), 2)
)
print("\nRecords per source:")
for source in sources:
count = sum(record["source"] == source for record in records)
print(source, ":", count)
# Assessment Data 5: Ordered Results
print("\n=== ASSESSMENT DATA 5: ORDERED RESULTS ===")
for record in sorted(organized, key=lambda r: (r["category"], r["value"])):
print(record)
# Assessment Data 6: Final Summary
print("\n=== FINAL OUTPUT: CONSOLIDATED TEST DATA SUMMARY ===")
print("Total records:", len(records))
print("Distinct source/category combinations:", len(seen))
print("Repeated records:", len(repeated))
print("Measurement categories:", len(categories))
print("Test sources:", len(sources))
print("All records have been generated, organized, and summarized.")
[1]
Python
=== ASSESSMENT DATA 1: GENERATED TEST DATA ===
{'source': 'ZARAGOZA_T1', 'category': 'TEMPERATURE', 'value': 87}
{'source': 'ZARAGOZA_T1', 'category': 'VOLTAGE', 'value': 9}
{'source': 'ZARAGOZA_T1', 'category': 'CURRENT', 'value': 9}
{'source': 'ZARAGOZA_T1', 'category': 'PRESSURE', 'value': 110}
{'source': 'ZARAGOZA_T1', 'category': 'SPEED', 'value': 19}
{'source': 'ZARAGOZA_T2', 'category': 'TEMPERATURE', 'value': 23}
{'source': 'ZARAGOZA_T2', 'category': 'VOLTAGE', 'value': 5}
{'source': 'ZARAGOZA_T2', 'category': 'CURRENT', 'value': 8}
Chat
Sessions
ValueError invalid literal for int
1 mo ago
Tip: Reference files or folders with # to give the agent more context about the task.
EXERCISE1.ipynb • Cell 2
Chat with Copilot
Agent
Copilot
Lab4_DataStructures_Zaragoza main*
0 4
Ln 122, Col 50
Spaces: 4
Cell 2 of 2

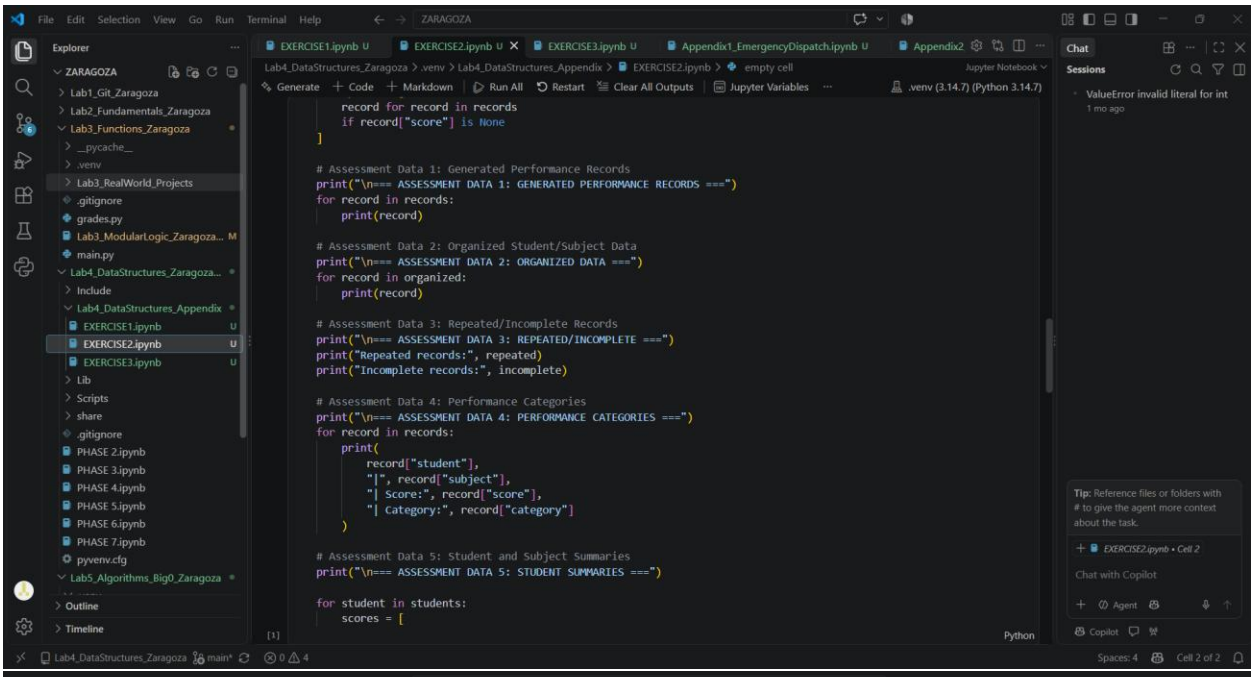


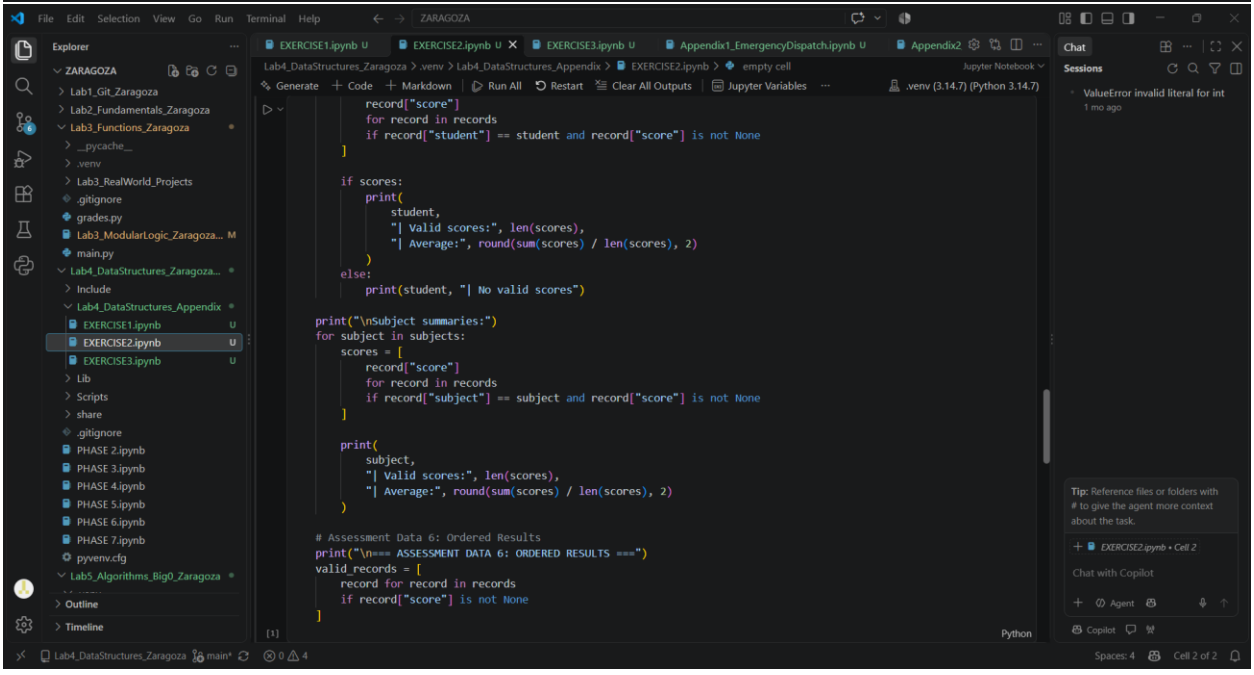

# EXERCISE 2: Student Performance Data Analyzer

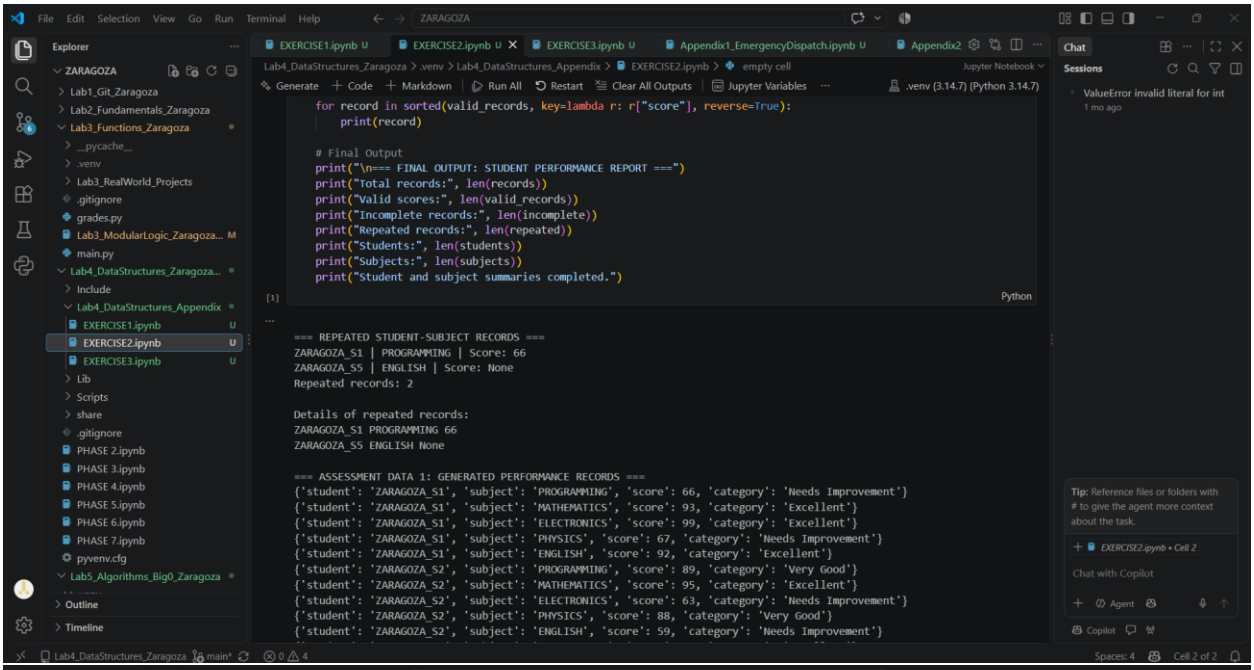


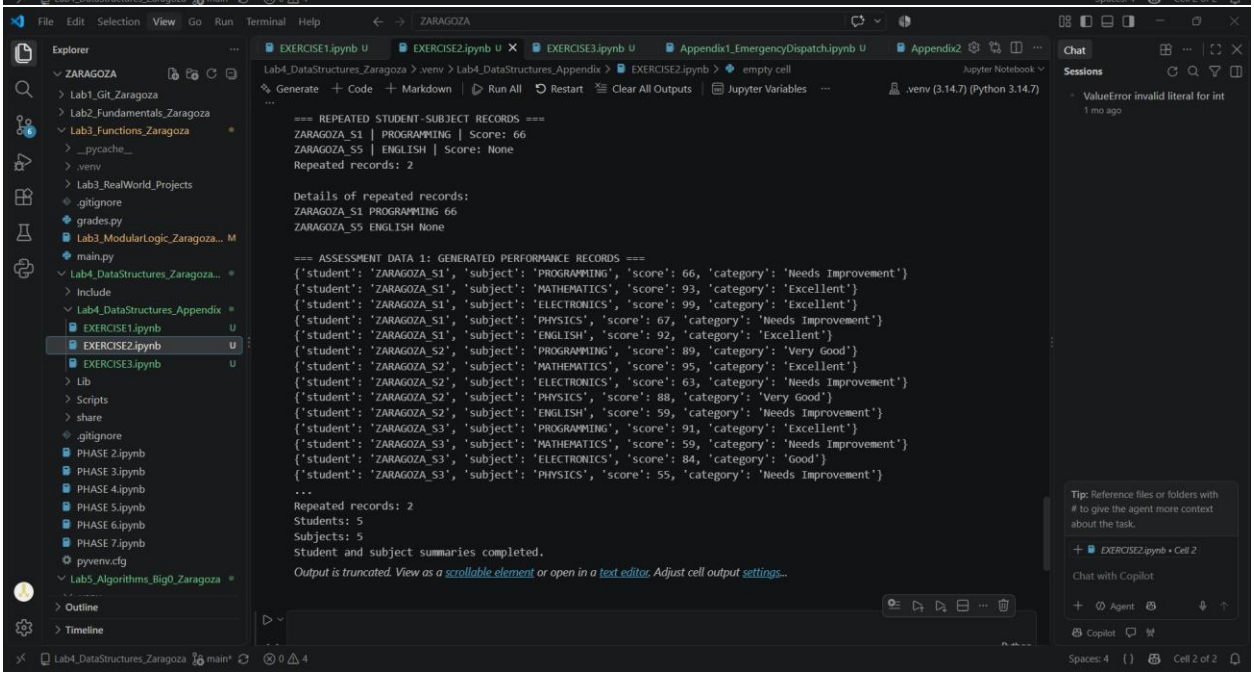

File Edit Selection View Go Run Terminal Help
ZARAGOZA
Explorer
EXERCISE1.ipynb U
EXERCISE2.ipynb U
EXERCISE3.ipynb U
Appendix1_EmergencyDispatch.ipynb U
Appendix2
Lab4_DataStructures_Zaragoza > .venv > Lab4_DataStructures_Appendix > EXERCISE2.ipynb > empty cell
Jupyter Notebook
Generate Code Markdown Run All Restart Clear All Outputs Jupyter Variables
.venv (3.14.7) (Python 3.14.7)
Chat
Sessions
ValueError invalid literal for int
1 mo ago
# Generate a score for each student and subject.
for student_index, student in enumerate(students):
for subject_index, subject in enumerate(subjects):
artist_value = artist_letters[
(student_index + subject_index) % len(artist_letters)
]
score = 50 + (
SEED_NUM
+ artist_value
+ student_index * 13
+ subject_index * 17
) % 51
records.append({
"student": student,
"subject": subject,
"score": score,
"category": performance_category(score)
})
# Add a repeated student/subject record.
records.append(records[0].copy())
# Add one incomplete record with a missing score.
records.append({
"student": students[4],
"subject": subjects[4],
"score": None,
"category": "MISSING"
})
# Organize records by student and subject.
organized = sorted(
[1]
Python
Tip: Reference files or folders with # to give the agent more context about the task.
EXERCISE2.ipynb • Cell 2
Chat with Copilot
Agent
Copilot
Lab4_DataStructures_Zaragoza main* 0 4
Spaces: 4
Cell 2 of 2


File Edit Selection View Go Run Terminal Help
ZARAGOZA
Explorer
EXERCISE1.ipynb U
EXERCISE2.ipynb U
EXERCISE3.ipynb U
Appendix1_EmergencyDispatch.ipynb U
Appendix2
Lab4_DataStructures_Zaragoza > .venv > Lab4_DataStructures_Appendix > EXERCISE2.ipynb > empty cell
Jupyter Notebook
Generate Code Markdown Run All Restart Clear All Outputs Jupyter Variables
.venv (3.14.7) (Python 3.14.7)
Chat
Sessions
ValueError invalid literal for int
1 mo ago
records,
key=lambda record: (record["student"], record["subject"])
)
# Find repeated student/subject records.
seen = set()
repeated = []
for record in organized:
key = (record["student"], record["subject"])
if key in seen:
repeated.append(record)
else:
seen.add(key)
print("\n=== REPEATED STUDENT-SUBJECT RECORDS ===")
for record in repeated:
print(
record["student"],
"|",
record["subject"],
"| Score:",
record["score"]
)
print("Repeated records:", len(repeated))
print("\nDetails of repeated records:")
for record in repeated:
print(record["student"], record["subject"], record["score"])
incomplete = [
[1]
Python
Tip: Reference files or folders with # to give the agent more context about the task.
EXERCISE2.ipynb • Cell 2
Chat with Copilot
Agent
Copilot
Lab4_DataStructures_Zaragoza main* 0 4
Spaces: 4
Cell 2 of 2

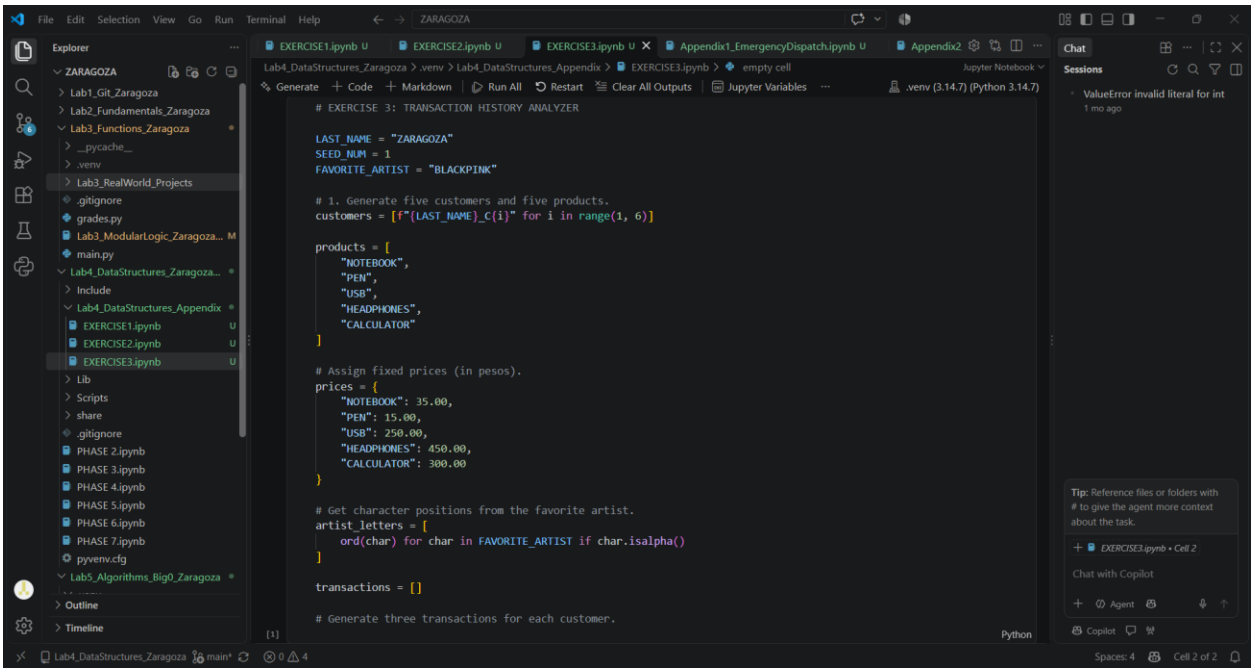

File Edit Selection View Go Run Terminal Help
ZARAGOZA
Explorer
EXERCISE1.ipynb U
EXERCISE2.ipynb U
EXERCISE3.ipynb U
Appendix1_EmergencyDispatch.ipynb U
Appendix2
Generate
Code
Markdown
Run All
Restart
Clear All Outputs
Jupyter Variables
.venv (3.14.7) (Python 3.14.7)
record for record in records
if record["score"] is None
# Assessment Data 1: Generated Performance Records
print("\n=== ASSESSMENT DATA 1: GENERATED PERFORMANCE RECORDS ===")
for record in records:
print(record)
# Assessment Data 2: Organized Student/Subject Data
print("\n=== ASSESSMENT DATA 2: ORGANIZED DATA ===")
for record in organized:
print(record)
# Assessment Data 3: Repeated/Incomplete Records
print("\n=== ASSESSMENT DATA 3: REPEATED/INCOMPLETE ===")
print("Repeated records:", repeated)
print("Incomplete records:", incomplete)
# Assessment Data 4: Performance Categories
print("\n=== ASSESSMENT DATA 4: PERFORMANCE CATEGORIES ===")
for record in records:
print(
record["student"],
"|", record["subject"],
"| Score:", record["score"],
"| Category:", record["category"]
)
# Assessment Data 5: Student and Subject Summaries
print("\n=== ASSESSMENT DATA 5: STUDENT SUMMARIES ===")
for student in students:
scores = [
Python
Chat
Sessions
ValueError invalid literal for int
1 mo ago
Tip: Reference files or folders with # to give the agent more context about the task.
Chat with Copilot
Agent
Copilot
Lab4_DataStructures_Zaragoza
main*
Spaces: 4
Cell 2 of 2


File Edit Selection View Go Run Terminal Help
ZARAGOZA
Explorer
EXERCISE1.ipynb U
EXERCISE2.ipynb U
EXERCISE3.ipynb U
Appendix1_EmergencyDispatch.ipynb U
Appendix2
Generate
Code
Markdown
Run All
Restart
Clear All Outputs
Jupyter Variables
.venv (3.14.7) (Python 3.14.7)
record["score"]
for record in records
if record["student"] == student and record["score"] is not None
if scores:
print(
student,
"| Valid scores:", len(scores),
"| Average:", round(sum(scores) / len(scores), 2)
)
else:
print(student, "| No valid scores")
print("\nSubject summaries:")
for subject in subjects:
scores = [
record["score"]
for record in records
if record["subject"] == subject and record["score"] is not None
]
print(
subject,
"| Valid scores:", len(scores),
"| Average:", round(sum(scores) / len(scores), 2)
)
# Assessment Data 6: Ordered Results
print("\n=== ASSESSMENT DATA 6: ORDERED RESULTS ===")
valid_records = [
record for record in records
if record["score"] is not None
]
Python
Chat
Sessions
ValueError invalid literal for int
1 mo ago
Tip: Reference files or folders with # to give the agent more context about the task.
Chat with Copilot
Agent
Copilot
Lab4_DataStructures_Zaragoza
main*
Spaces: 4
Cell 2 of 2


File Edit Selection View Go Run Terminal Help
ZARAGOZA
Explorer
ZARAGOZA
Lab1_Git_Zaragoza
Lab2_Fundamentals_Zaragoza
Lab3_Functions_Zaragoza
__pycache__
.venv
Lab3_RealWorld_Projects
.gitignore
grades.py
Lab3_ModularLogic_Zaragoza... M
main.py
Lab4_DataStructures_Zaragoza...
Include
Lab4_DataStructures_Appendix
EXERCISE1.ipynb U
EXERCISE2.ipynb U
EXERCISE3.ipynb U
Lib
Scripts
share
.gitignore
PHASE 2.ipynb
PHASE 3.ipynb
PHASE 4.ipynb
PHASE 5.ipynb
PHASE 6.ipynb
PHASE 7.ipynb
pyvenv.cfg
Lab5_Algorithms_Big0_Zaragoza
Outline
Timeline
EXERCISE1.ipynb U
EXERCISE2.ipynb U
EXERCISE3.ipynb U
Appendix1_EmergencyDispatch.ipynb U
Appendix2
Lab4_DataStructures_Zaragoza > .venv > Lab4_DataStructures_Appendix > EXERCISE2.ipynb > empty cell
Jupyter Notebook
Generate Code Markdown Run All Restart Clear All Outputs Jupyter Variables
.venv (3.14.7) (Python 3.14.7)
for record in sorted(valid_records, key=lambda r: r["score"], reverse=True):
print(record)
# Final Output
print("\n=== FINAL OUTPUT: STUDENT PERFORMANCE REPORT ===")
print("Total records:", len(records))
print("Valid scores:", len(valid_records))
print("Incomplete records:", len(incomplete))
print("Repeated records:", len(repeated))
print("Students:", len(students))
print("Subjects:", len(subjects))
print("Student and subject summaries completed.")
[1]
Python
=== REPEATED STUDENT-SUBJECT RECORDS ===
ZARAGOZA_S1 | PROGRAMMING | Score: 66
ZARAGOZA_S5 | ENGLISH | Score: None
Repeated records: 2
Details of repeated records:
ZARAGOZA_S1 PROGRAMMING 66
ZARAGOZA_S5 ENGLISH None
=== ASSESSMENT DATA 1: GENERATED PERFORMANCE RECORDS ===
{'student': 'ZARAGOZA_S1', 'subject': 'PROGRAMMING', 'score': 66, 'category': 'Needs Improvement'}
{'student': 'ZARAGOZA_S1', 'subject': 'MATHEMATICS', 'score': 93, 'category': 'Excellent'}
{'student': 'ZARAGOZA_S1', 'subject': 'ELECTRONICS', 'score': 99, 'category': 'Excellent'}
{'student': 'ZARAGOZA_S1', 'subject': 'PHYSICS', 'score': 67, 'category': 'Needs Improvement'}
{'student': 'ZARAGOZA_S1', 'subject': 'ENGLISH', 'score': 92, 'category': 'Excellent'}
{'student': 'ZARAGOZA_S2', 'subject': 'PROGRAMMING', 'score': 89, 'category': 'Very Good'}
{'student': 'ZARAGOZA_S2', 'subject': 'MATHEMATICS', 'score': 95, 'category': 'Excellent'}
{'student': 'ZARAGOZA_S2', 'subject': 'ELECTRONICS', 'score': 63, 'category': 'Needs Improvement'}
{'student': 'ZARAGOZA_S2', 'subject': 'PHYSICS', 'score': 88, 'category': 'Very Good'}
{'student': 'ZARAGOZA_S2', 'subject': 'ENGLISH', 'score': 59, 'category': 'Needs Improvement'}
Chat
Sessions
ValueError invalid literal for int
1 mo ago
Tip: Reference files or folders with # to give the agent more context about the task.
EXERCISE2.ipynb • Cell 2
Chat with Copilot
Agent
Copilot
Lab4_DataStructures_Zaragoza main*
0 4
Spaces: 4
Cell 2 of 2


File Edit Selection View Go Run Terminal Help
ZARAGOZA
Explorer
ZARAGOZA
Lab1_Git_Zaragoza
Lab2_Fundamentals_Zaragoza
Lab3_Functions_Zaragoza
__pycache__
.venv
Lab3_RealWorld_Projects
.gitignore
grades.py
Lab3_ModularLogic_Zaragoza... M
main.py
Lab4_DataStructures_Zaragoza...
Include
Lab4_DataStructures_Appendix
EXERCISE1.ipynb U
EXERCISE2.ipynb U
EXERCISE3.ipynb U
Lib
Scripts
share
.gitignore
PHASE 2.ipynb
PHASE 3.ipynb
PHASE 4.ipynb
PHASE 5.ipynb
PHASE 6.ipynb
PHASE 7.ipynb
pyvenv.cfg
Lab5_Algorithms_Big0_Zaragoza
Outline
Timeline
EXERCISE1.ipynb U
EXERCISE2.ipynb U
EXERCISE3.ipynb U
Appendix1_EmergencyDispatch.ipynb U
Appendix2
Lab4_DataStructures_Zaragoza > .venv > Lab4_DataStructures_Appendix > EXERCISE2.ipynb > empty cell
Jupyter Notebook
Generate Code Markdown Run All Restart Clear All Outputs Jupyter Variables
.venv (3.14.7) (Python 3.14.7)
=== REPEATED STUDENT-SUBJECT RECORDS ===
ZARAGOZA_S1 | PROGRAMMING | Score: 66
ZARAGOZA_S5 | ENGLISH | Score: None
Repeated records: 2
Details of repeated records:
ZARAGOZA_S1 PROGRAMMING 66
ZARAGOZA_S5 ENGLISH None
=== ASSESSMENT DATA 1: GENERATED PERFORMANCE RECORDS ===
{'student': 'ZARAGOZA_S1', 'subject': 'PROGRAMMING', 'score': 66, 'category': 'Needs Improvement'}
{'student': 'ZARAGOZA_S1', 'subject': 'MATHEMATICS', 'score': 93, 'category': 'Excellent'}
{'student': 'ZARAGOZA_S1', 'subject': 'ELECTRONICS', 'score': 99, 'category': 'Excellent'}
{'student': 'ZARAGOZA_S1', 'subject': 'PHYSICS', 'score': 67, 'category': 'Needs Improvement'}
{'student': 'ZARAGOZA_S1', 'subject': 'ENGLISH', 'score': 92, 'category': 'Excellent'}
{'student': 'ZARAGOZA_S2', 'subject': 'PROGRAMMING', 'score': 89, 'category': 'Very Good'}
{'student': 'ZARAGOZA_S2', 'subject': 'MATHEMATICS', 'score': 95, 'category': 'Excellent'}
{'student': 'ZARAGOZA_S2', 'subject': 'ELECTRONICS', 'score': 63, 'category': 'Needs Improvement'}
{'student': 'ZARAGOZA_S2', 'subject': 'PHYSICS', 'score': 88, 'category': 'Very Good'}
{'student': 'ZARAGOZA_S2', 'subject': 'ENGLISH', 'score': 59, 'category': 'Needs Improvement'}
{'student': 'ZARAGOZA_S3', 'subject': 'PROGRAMMING', 'score': 91, 'category': 'Excellent'}
{'student': 'ZARAGOZA_S3', 'subject': 'MATHEMATICS', 'score': 59, 'category': 'Needs Improvement'}
{'student': 'ZARAGOZA_S3', 'subject': 'ELECTRONICS', 'score': 84, 'category': 'Good'}
{'student': 'ZARAGOZA_S3', 'subject': 'PHYSICS', 'score': 55, 'category': 'Needs Improvement'}
...
Repeated records: 2
Students: 5
Subjects: 5
Student and subject summaries completed.
Output is truncated. View as a scrollable element or open in a text editor. Adjust cell output settings...
Chat
Sessions
ValueError invalid literal for int
1 mo ago
Tip: Reference files or folders with # to give the agent more context about the task.
EXERCISE2.ipynb • Cell 2
Chat with Copilot
Agent
Copilot
Lab4_DataStructures_Zaragoza main*
0 4
Spaces: 4
Cell 2 of 2

# # EXERCISE 3: Transaction History Analyzer



```
# EXERCISE 3: TRANSACTION HISTORY ANALYZER

LAST_NAME = "ZARAGOZA"
SEED_NUM = 1
FAVORITE_ARTIST = "BLACKPINK"

# 1. Generate five customers and five products.
customers = [f"{LAST_NAME}_C{i}" for i in range(1, 6)]

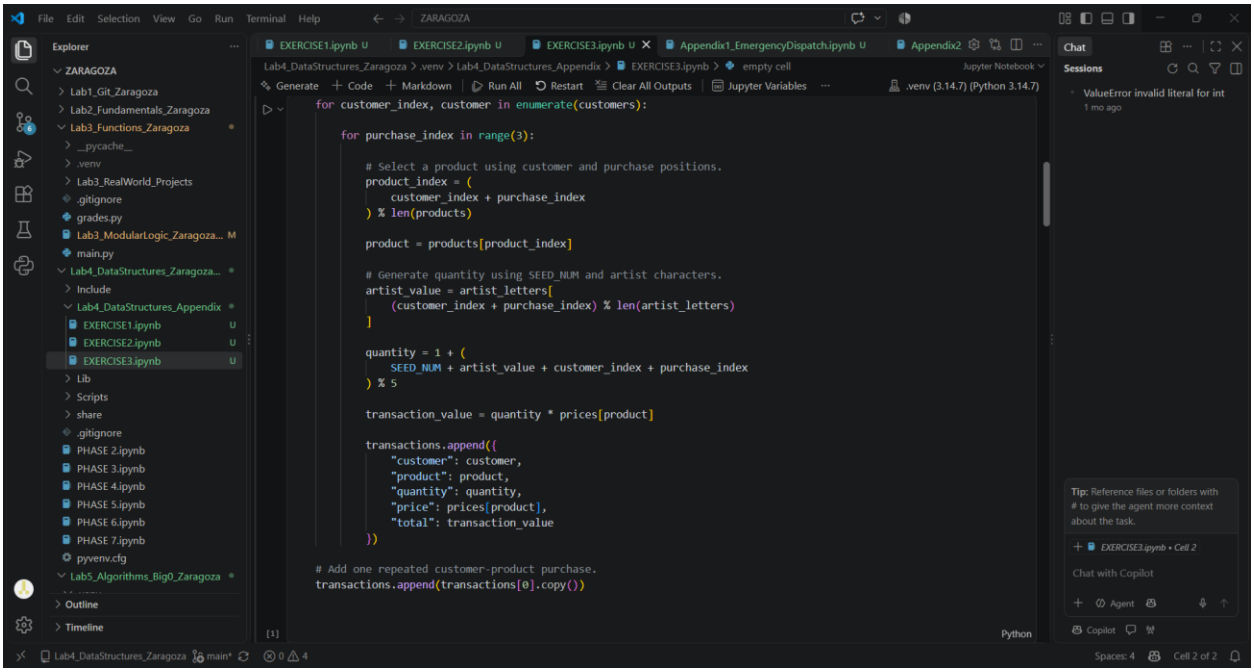
products = [
    "NOTEBOOK",
    "PEN",
    "USB",
    "HEADPHONES",
    "CALCULATOR"
]

# Assign fixed prices (in pesos).
prices = {
    "NOTEBOOK": 35.00,
    "PEN": 15.00,
    "USB": 250.00,
    "HEADPHONES": 450.00,
    "CALCULATOR": 300.00
}

# Get character positions from the favorite artist.
artist_letters = [
    ord(char) for char in FAVORITE_ARTIST if char.isalpha()
]

transactions = []

# Generate three transactions for each customer.
```



```
for customer_index, customer in enumerate(customers):

    for purchase_index in range(3):

        # Select a product using customer and purchase positions.
        product_index = (
            customer_index + purchase_index
        ) % len(products)

        product = products[product_index]

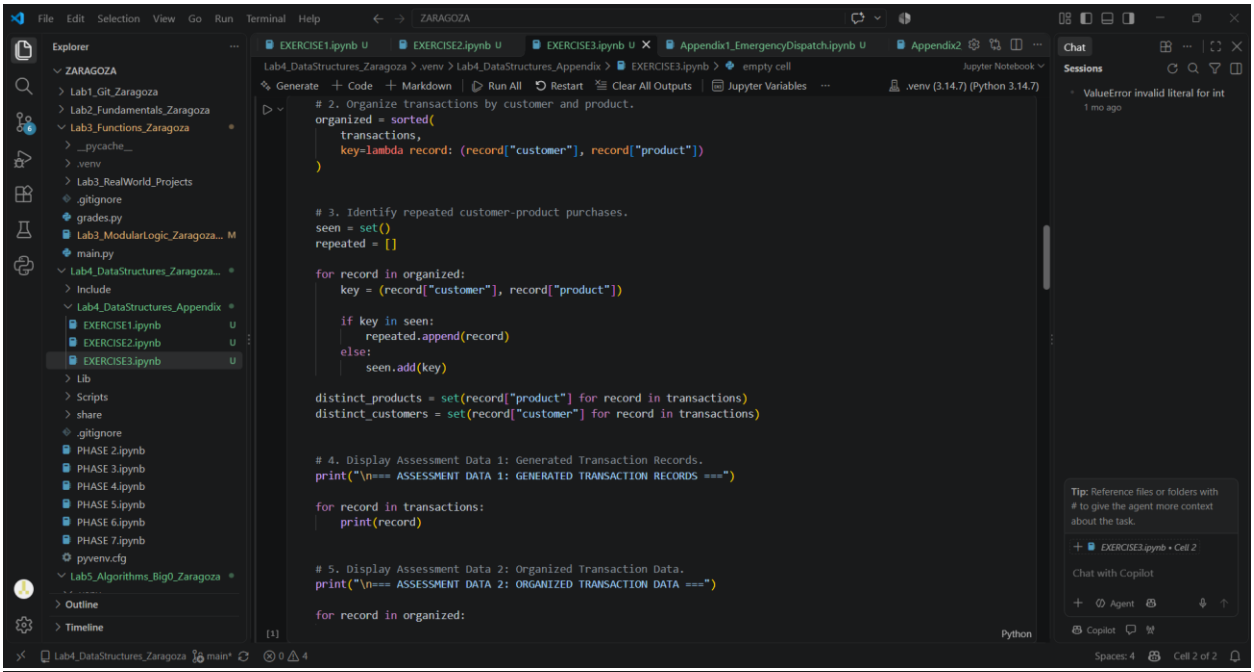
        # Generate quantity using SEED_NUM and artist characters.
        artist_value = artist_letters[
            (customer_index + purchase_index) % len(artist_letters)
        ]

        quantity = 1 + (
            SEED_NUM + artist_value + customer_index + purchase_index
        ) % 5

        transaction_value = quantity * prices[product]

        transactions.append({
            "customer": customer,
            "product": product,
            "quantity": quantity,
            "price": prices[product],
            "total": transaction_value
        })

# Add one repeated customer-product purchase.
transactions.append(transactions[0].copy())
```



```
# 2. Organize transactions by customer and product.
organized = sorted(
    transactions,
    key=lambda record: (record["customer"], record["product"])
)


# 3. Identify repeated customer-product purchases.
seen = set()
repeated = []

for record in organized:
    key = (record["customer"], record["product"])

    if key in seen:
        repeated.append(record)
    else:
        seen.add(key)

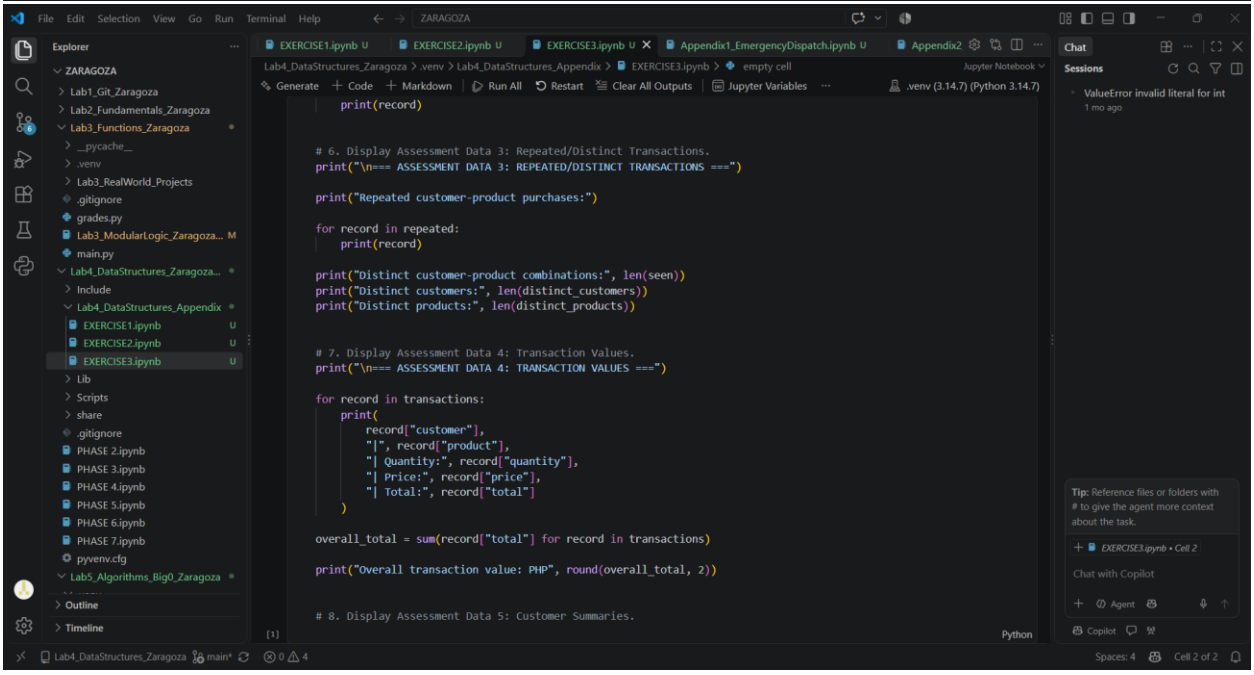
distinct_products = set(record["product"] for record in transactions)
distinct_customers = set(record["customer"] for record in transactions)


# 4. Display Assessment Data 1: Generated Transaction Records.
print("\n=== ASSESSMENT DATA 1: GENERATED TRANSACTION RECORDS ===")

for record in transactions:
    print(record)


# 5. Display Assessment Data 2: Organized Transaction Data.
print("\n=== ASSESSMENT DATA 2: ORGANIZED TRANSACTION DATA ===")

for record in organized:
```



```
    print(record)


# 6. Display Assessment Data 3: Repeated/Distinct Transactions.
print("\n=== ASSESSMENT DATA 3: REPEATED/DISTINCT TRANSACTIONS ===")

print("Repeated customer-product purchases:")

for record in repeated:
    print(record)

print("Distinct customer-product combinations:", len(seen))
print("Distinct customers:", len(distinct_customers))
print("Distinct products:", len(distinct_products))

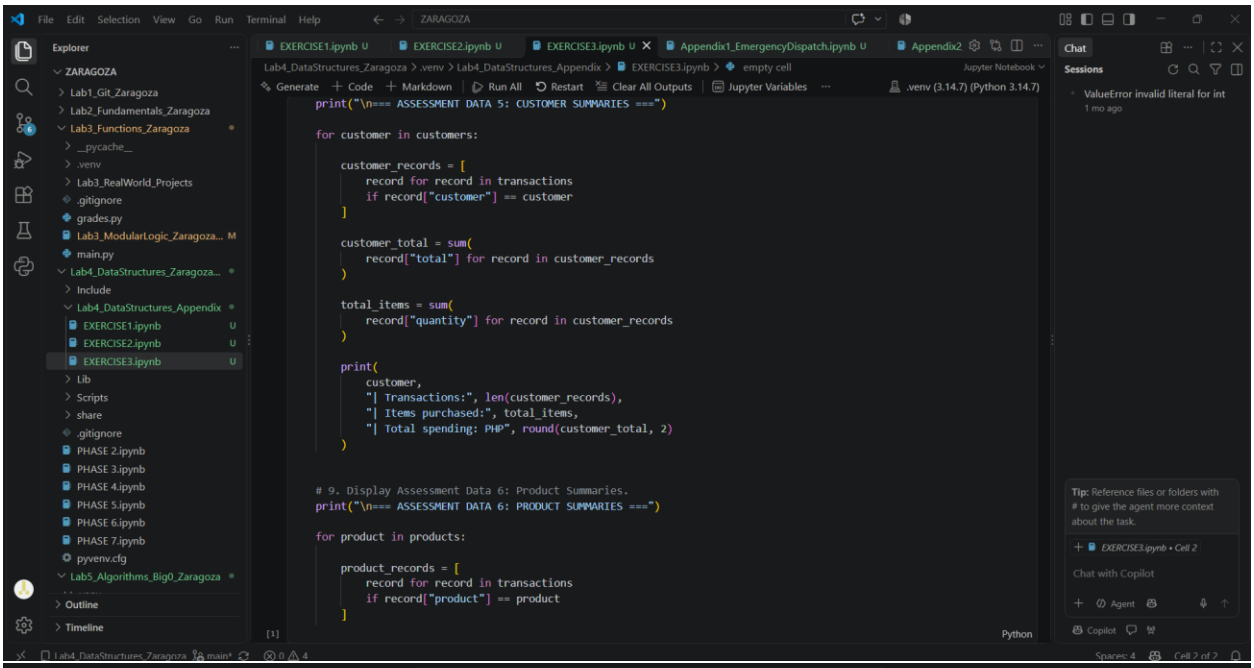

# 7. Display Assessment Data 4: Transaction Values.
print("\n=== ASSESSMENT DATA 4: TRANSACTION VALUES ===")

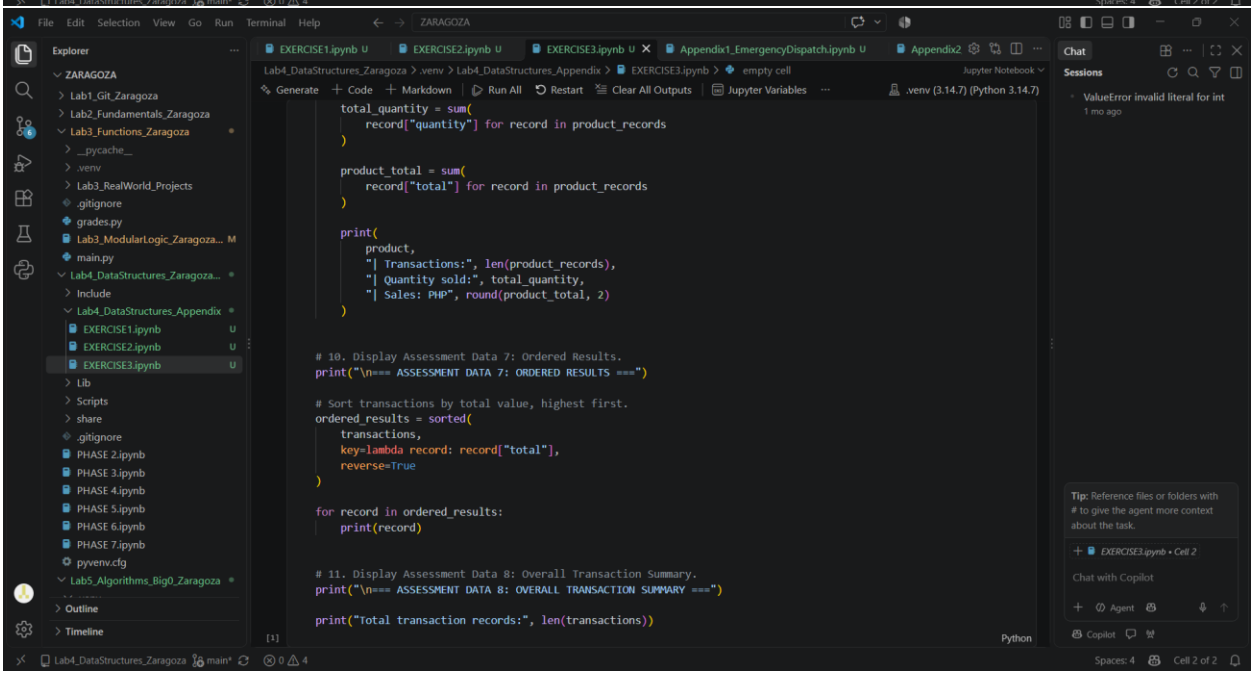
for record in transactions:
    print(
        record["customer"],
        "|", record["product"],
        "| Quantity:", record["quantity"],
        "| Price:", record["price"],
        "| Total:", record["total"]
    )

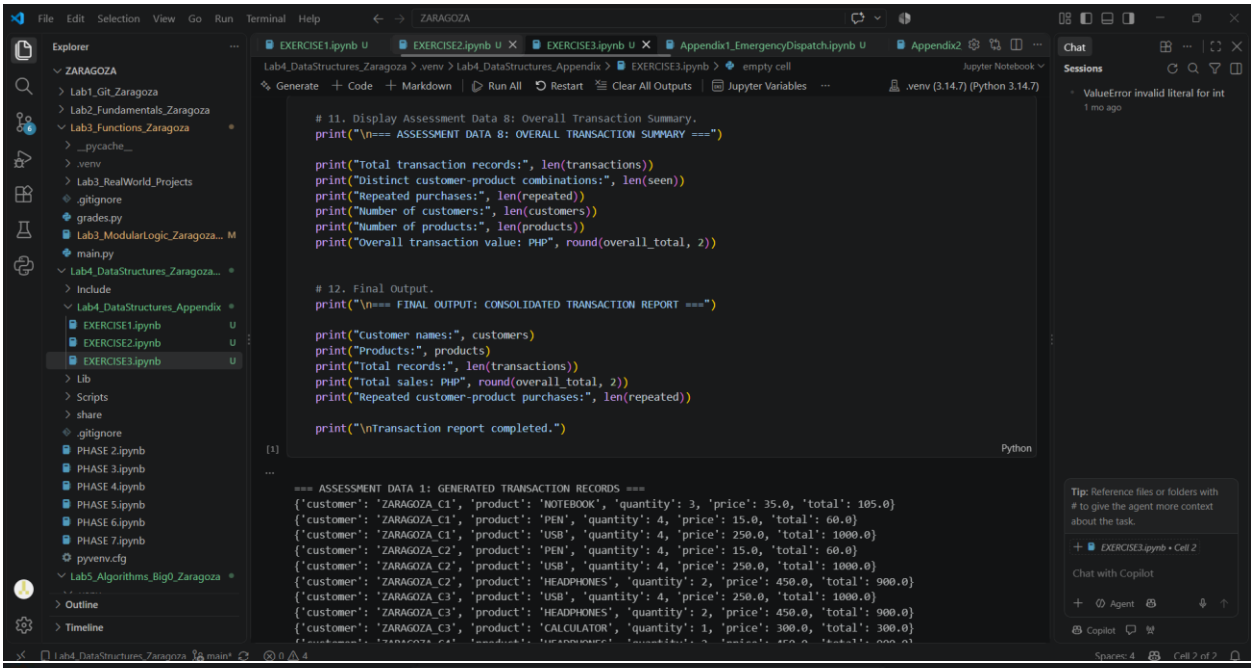
overall_total = sum(record["total"] for record in transactions)

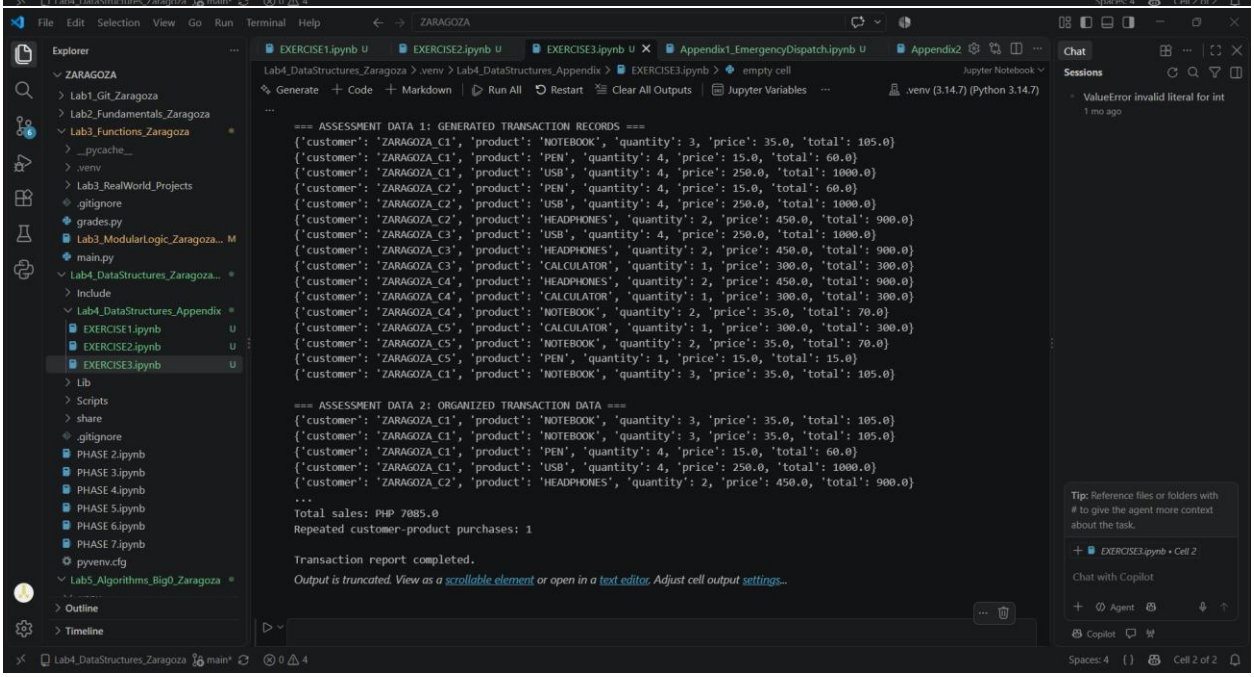
print("Overall transaction value: PHP", round(overall_total, 2))


# 8. Display Assessment Data 5: Customer Summaries.
```


File Edit Selection View Go Run Terminal Help
ZARAGOZA
Explorer
EXERCISE1.ipynb U
EXERCISE2.ipynb U
EXERCISE3.ipynb U
Appendix1_EmergencyDispatch.ipynb U
Appendix2
Generate Code Markdown Run All Restart Clear All Outputs Jupyter Variables
.venv (3.14.7) (Python 3.14.7)
print("\n=== ASSESSMENT DATA 5: CUSTOMER SUMMARIES ===")
for customer in customers:
customer_records = [
record for record in transactions
if record["customer"] == customer
]
customer_total = sum(
record["total"] for record in customer_records
)
total_items = sum(
record["quantity"] for record in customer_records
)
print(
customer,
"| Transactions:", len(customer_records),
"| Items purchased:", total_items,
"| Total spending: PHP", round(customer_total, 2)
)
# 9. Display Assessment Data 6: Product Summaries.
print("\n=== ASSESSMENT DATA 6: PRODUCT SUMMARIES ===")
for product in products:
product_records = [
record for record in transactions
if record["product"] == product
]
Chat
Sessions
ValueError invalid literal for int
1 mo ago
Tip: Reference files or folders with # to give the agent more context about the task.
Chat with Copilot
Agent
Copilot
Python
Spaces: 4
Cell 2 of 2


File Edit Selection View Go Run Terminal Help
ZARAGOZA
Explorer
EXERCISE1.ipynb U
EXERCISE2.ipynb U
EXERCISE3.ipynb U
Appendix1_EmergencyDispatch.ipynb U
Appendix2
Generate Code Markdown Run All Restart Clear All Outputs Jupyter Variables
.venv (3.14.7) (Python 3.14.7)
total_quantity = sum(
record["quantity"] for record in product_records
)
product_total = sum(
record["total"] for record in product_records
)
print(
product,
"| Transactions:", len(product_records),
"| Quantity sold:", total_quantity,
"| Sales: PHP", round(product_total, 2)
)
# 10. Display Assessment Data 7: Ordered Results.
print("\n=== ASSESSMENT DATA 7: ORDERED RESULTS ===")
# Sort transactions by total value, highest first.
ordered_results = sorted(
transactions,
key=lambda record: record["total"],
reverse=True
)
for record in ordered_results:
print(record)
# 11. Display Assessment Data 8: Overall Transaction Summary.
print("\n=== ASSESSMENT DATA 8: OVERALL TRANSACTION SUMMARY ===")
print("Total transaction records:", len(transactions))
Chat
Sessions
ValueError invalid literal for int
1 mo ago
Tip: Reference files or folders with # to give the agent more context about the task.
Chat with Copilot
Agent
Copilot
Python
Spaces: 4
Cell 2 of 2


File Edit Selection View Go Run Terminal Help
ZARAGOZA
Explorer
ZARAGOZA
Lab1_Git_Zaragoza
Lab2_Fundamentals_Zaragoza
Lab3_Functions_Zaragoza
__pycache__
.venv
Lab3_RealWorld_Projects
.gitignore
grades.py
Lab3_ModularLogic_Zaragoza... M
main.py
Lab4_DataStructures_Zaragoza...
Include
Lab4_DataStructures_Appendix
EXERCISE1.ipynb U
EXERCISE2.ipynb U
EXERCISE3.ipynb U
Lib
Scripts
share
.gitignore
PHASE 2.ipynb
PHASE 3.ipynb
PHASE 4.ipynb
PHASE 5.ipynb
PHASE 6.ipynb
PHASE 7.ipynb
pyvenv.cfg
Lab5_Algorithms_Big0_Zaragoza
Outline
Timeline
EXERCISE1.ipynb U
EXERCISE2.ipynb U
EXERCISE3.ipynb U
Appendix1_EmergencyDispatch.ipynb U
Appendix2
Lab4_DataStructures_Zaragoza > .venv > Lab4_DataStructures_Appendix > EXERCISE3.ipynb > empty cell
Jupyter Notebook
Generate
Code
Markdown
Run All
Restart
Clear All Outputs
Jupyter Variables
.venv (3.14.7) (Python 3.14.7)
# 11. Display Assessment Data 8: Overall Transaction Summary.
print("\n=== ASSESSMENT DATA 8: OVERALL TRANSACTION SUMMARY ===")
print("Total transaction records:", len(transactions))
print("Distinct customer-product combinations:", len(seen))
print("Repeated purchases:", len(repeated))
print("Number of customers:", len(customers))
print("Number of products:", len(products))
print("Overall transaction value: PHP", round(overall_total, 2))
# 12. Final Output.
print("\n=== FINAL OUTPUT: CONSOLIDATED TRANSACTION REPORT ===")
print("Customer names:", customers)
print("Products:", products)
print("Total records:", len(transactions))
print("Total sales: PHP", round(overall_total, 2))
print("Repeated customer-product purchases:", len(repeated))
print("\nTransaction report completed.")
[1]
Python
=== ASSESSMENT DATA 1: GENERATED TRANSACTION RECORDS ===
{'customer': 'ZARAGOZA_C1', 'product': 'NOTEBOOK', 'quantity': 3, 'price': 35.0, 'total': 105.0}
{'customer': 'ZARAGOZA_C1', 'product': 'PEN', 'quantity': 4, 'price': 15.0, 'total': 60.0}
{'customer': 'ZARAGOZA_C1', 'product': 'USB', 'quantity': 4, 'price': 250.0, 'total': 1000.0}
{'customer': 'ZARAGOZA_C2', 'product': 'PEN', 'quantity': 4, 'price': 15.0, 'total': 60.0}
{'customer': 'ZARAGOZA_C2', 'product': 'USB', 'quantity': 4, 'price': 250.0, 'total': 1000.0}
{'customer': 'ZARAGOZA_C2', 'product': 'HEADPHONES', 'quantity': 2, 'price': 450.0, 'total': 900.0}
{'customer': 'ZARAGOZA_C3', 'product': 'USB', 'quantity': 4, 'price': 250.0, 'total': 1000.0}
{'customer': 'ZARAGOZA_C3', 'product': 'HEADPHONES', 'quantity': 2, 'price': 450.0, 'total': 900.0}
{'customer': 'ZARAGOZA_C3', 'product': 'CALCULATOR', 'quantity': 1, 'price': 300.0, 'total': 300.0}
Chat
Sessions
ValueError invalid literal for int
1 mo ago
Tip: Reference files or folders with # to give the agent more context about the task.
EXERCISE3.ipynb • Cell 2
Chat with Copilot
Agent
Copilot
Lab4_DataStructures_Zaragoza main*
0 4
Spaces: 4
Cell 2 of 2


File Edit Selection View Go Run Terminal Help
ZARAGOZA
Explorer
ZARAGOZA
Lab1_Git_Zaragoza
Lab2_Fundamentals_Zaragoza
Lab3_Functions_Zaragoza
__pycache__
.venv
Lab3_RealWorld_Projects
.gitignore
grades.py
Lab3_ModularLogic_Zaragoza... M
main.py
Lab4_DataStructures_Zaragoza...
Include
Lab4_DataStructures_Appendix
EXERCISE1.ipynb U
EXERCISE2.ipynb U
EXERCISE3.ipynb U
Lib
Scripts
share
.gitignore
PHASE 2.ipynb
PHASE 3.ipynb
PHASE 4.ipynb
PHASE 5.ipynb
PHASE 6.ipynb
PHASE 7.ipynb
pyvenv.cfg
Lab5_Algorithms_Big0_Zaragoza
Outline
Timeline
EXERCISE1.ipynb U
EXERCISE2.ipynb U
EXERCISE3.ipynb U
Appendix1_EmergencyDispatch.ipynb U
Appendix2
Lab4_DataStructures_Zaragoza > .venv > Lab4_DataStructures_Appendix > EXERCISE3.ipynb > empty cell
Jupyter Notebook
Generate
Code
Markdown
Run All
Restart
Clear All Outputs
Jupyter Variables
.venv (3.14.7) (Python 3.14.7)
=== ASSESSMENT DATA 1: GENERATED TRANSACTION RECORDS ===
{'customer': 'ZARAGOZA_C1', 'product': 'NOTEBOOK', 'quantity': 3, 'price': 35.0, 'total': 105.0}
{'customer': 'ZARAGOZA_C1', 'product': 'PEN', 'quantity': 4, 'price': 15.0, 'total': 60.0}
{'customer': 'ZARAGOZA_C1', 'product': 'USB', 'quantity': 4, 'price': 250.0, 'total': 1000.0}
{'customer': 'ZARAGOZA_C2', 'product': 'PEN', 'quantity': 4, 'price': 15.0, 'total': 60.0}
{'customer': 'ZARAGOZA_C2', 'product': 'USB', 'quantity': 4, 'price': 250.0, 'total': 1000.0}
{'customer': 'ZARAGOZA_C2', 'product': 'HEADPHONES', 'quantity': 2, 'price': 450.0, 'total': 900.0}
{'customer': 'ZARAGOZA_C3', 'product': 'USB', 'quantity': 4, 'price': 250.0, 'total': 1000.0}
{'customer': 'ZARAGOZA_C3', 'product': 'HEADPHONES', 'quantity': 2, 'price': 450.0, 'total': 900.0}
{'customer': 'ZARAGOZA_C3', 'product': 'CALCULATOR', 'quantity': 1, 'price': 300.0, 'total': 300.0}
{'customer': 'ZARAGOZA_C4', 'product': 'HEADPHONES', 'quantity': 2, 'price': 450.0, 'total': 900.0}
{'customer': 'ZARAGOZA_C4', 'product': 'CALCULATOR', 'quantity': 1, 'price': 300.0, 'total': 300.0}
{'customer': 'ZARAGOZA_C4', 'product': 'NOTEBOOK', 'quantity': 2, 'price': 35.0, 'total': 70.0}
{'customer': 'ZARAGOZA_C5', 'product': 'CALCULATOR', 'quantity': 1, 'price': 300.0, 'total': 300.0}
{'customer': 'ZARAGOZA_C5', 'product': 'NOTEBOOK', 'quantity': 2, 'price': 35.0, 'total': 70.0}
{'customer': 'ZARAGOZA_C5', 'product': 'PEN', 'quantity': 1, 'price': 15.0, 'total': 15.0}
{'customer': 'ZARAGOZA_C1', 'product': 'NOTEBOOK', 'quantity': 3, 'price': 35.0, 'total': 105.0}
=== ASSESSMENT DATA 2: ORGANIZED TRANSACTION DATA ===
{'customer': 'ZARAGOZA_C1', 'product': 'NOTEBOOK', 'quantity': 3, 'price': 35.0, 'total': 105.0}
{'customer': 'ZARAGOZA_C1', 'product': 'NOTEBOOK', 'quantity': 3, 'price': 35.0, 'total': 105.0}
{'customer': 'ZARAGOZA_C1', 'product': 'PEN', 'quantity': 4, 'price': 15.0, 'total': 60.0}
{'customer': 'ZARAGOZA_C1', 'product': 'USB', 'quantity': 4, 'price': 250.0, 'total': 1000.0}
{'customer': 'ZARAGOZA_C2', 'product': 'HEADPHONES', 'quantity': 2, 'price': 450.0, 'total': 900.0}
...
Total sales: PHP 7085.0
Repeated customer-product purchases: 1
Transaction report completed.
Output is truncated. View as a scrollable element or open in a text editor. Adjust cell output settings...
Chat
Sessions
ValueError invalid literal for int
1 mo ago
Tip: Reference files or folders with # to give the agent more context about the task.
EXERCISE3.ipynb • Cell 2
Chat with Copilot
Agent
Copilot
Lab4_DataStructures_Zaragoza main*
0 4
Spaces: 4
Cell 2 of 2